WORKED SOLUTION

STEP II 2009, Question 9

Mechanics • Centre of Mass

STEP II 2009, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q9

The key idea

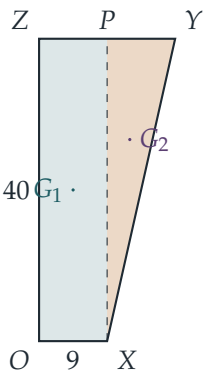
Every distance we need is measured perpendicular to one fixed plane: the edge OZ in part (i), and the back of the tank in part (ii), which contains OZ. So we only ever need one coordinate. Split each shape into pieces whose centres of mass we know (rectangles and a triangle), take moments about that line or plane, and divide by the total mass. The tank is the same calculation one level up: its pieces are two trapezia and three rectangles of sheet metal, and then a block of liquid.

Solution

Part (i)

Take O as the origin, with the x-axis along OX and the y-axis along OZ. Then $X = (9, 0)$, $Z = (0, 40)$ and, since YZ is parallel to OX with length 18, $Y = (18, 40)$. As a check, $XY^2 = 9^2 + 40^2 = 1681 = 41^2$, as it should be. The distance from OZ is just the x-coordinate.

Draw the line from X parallel to OZ, meeting ZY at $P = (9, 40)$. This splits the lamina into a rectangle $OXPZ$ and a right-angled triangle XYP.



Let the lamina have mass 1 per cm^2, so mass equals area.

- Rectangle $OXPZ$: area $9 \times 40 = 360$, centre G_1 at $x = 4.5$.
- Triangle XYP: area $\frac{1}{2} \times 9 \times 40 = 180$. The centre of mass of a uniform triangle is at the mean of its vertices, so G_2 has $x = \frac{1}{3}(9 + 18 + 9) = 12$.

The whole lamina has area 540. Taking moments about OZ,

$$540\,\bar{x} = 360 \times 4.5 + 180 \times 12 = 1620 + 2160 = 3780,$$

so $\bar{x} = 7$. The centre of mass is 7 cm from OZ, as required.

Part (ii)

Now let the sheet metal have mass 1 per cm^2. The back $OTWZ$ lies in the plane $x = 0$, and every point of the tank keeps the x-coordinate it had in part (i), since the width d runs perpendicular to the end faces. So we take moments about the back. Each rectangular face is uniform, so its centre of mass is at its centre, which has the same x-coordinate as the midpoint of the corresponding edge of the trapezium.

Piece	Area (cm^2)	x of centre (cm)	Moment
Two ends	$2 \times 540 = 1080$	7	7560
Base $OXUT$	$9d$	4.5	$40.5d$
Back $OTWZ$	$40d$	0	0
Front $XUVY$	$41d$	$\frac{1}{2}(9 + 18) = 13.5$	$553.5d$
Total	$1080 + 90d$		$7560 + 594d$

Ask yourself Where is the centre of the sloping front face?

The front is a rectangle with one pair of edges XU and YV, which lie at $x = 9$ and $x = 18$. Its centre is halfway between them, at $x = 13.5$. Its area uses the slant length $XY = 41$, not the horizontal width 9.

So the empty tank has

$$\bar{x} = \frac{7560 + 594d}{1080 + 90d}.$$

Every coefficient is divisible by 18: $7560 = 18 \times 420$, $594 = 18 \times 33$, $1080 = 18 \times 60$ and $90 = 18 \times 5$. So

$$\bar{x} = \frac{420 + 33d}{60 + 5d} = \frac{3(140 + 11d)}{5(12 + d)} \text{ cm},$$

which is the required distance from the back.

The filled tank

The liquid fills a prism with cross-section the trapezium of part (i) and length d. Its volume is $540d$, so its mass is $540dk$ in our units (mass per unit volume is k times the

metal's mass per unit area, which we took to be 1). Every cross-section is the same trapezium, so the liquid's centre of mass is at $x = 7$, as in part (i).

With $d = 20$, the tank has mass $1080 + 1800 = 2880$ and moment $7560 + 594 \times 20 = 19440$ about the back. The liquid has mass $10800k$ and moment $75600k$. So the filled tank has

$$\bar{x} = \frac{19440 + 75600k}{2880 + 10800k} = \frac{27 + 105k}{4 + 15k} \text{ cm},$$

after dividing top and bottom by 720. This is the distance of the centre of mass of the filled tank from the back.

Checks

- With $k = 0$ (no liquid, or liquid of negligible density) the answer is $\frac{27}{4} = 6.75$, and the empty-tank formula at $d = 20$ gives $\frac{3\times360}{5\times32} = 6.75$ too.
- As $k \to \infty$ the liquid dominates and $\bar{x} \to \frac{105}{15} = 7$, the liquid's own centre of mass.
- The answer always lies between 6.75 and 7, as a weighted average of the two must. For example, $k = 1$ gives $\frac{132}{19} \approx 6.947$.
- A numerical check (shoelace centroid of the trapezium, a Monte Carlo estimate giving $\bar{x} \approx 6.9994$, and exact fractions for both tank formulae) agrees with all the results.

Common slips

- Using $9d$ instead of $41d$ for the area of the front: the face is sloping, so its width is the slant length XY.
- Leaving out the back because its moment is zero. It still contributes $40d$ to the total mass.
- Counting only one end, or giving the tank a lid.

Beyond STEP

Here we found centres of mass by splitting into pieces with known centres. At university, the centre of mass of a lamina or solid is defined by integrals such as $\bar{x} = \frac{1}{M}\iint x\,\rho\,\mathrm{d}A$, and the "split into pieces" method is just the fact that integrals add over a region cut into parts. The same integrals give moments of inertia, which control how bodies rotate. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

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